

SUPPORTLYTICS

Optimizing IT Support Team Performance Using Analytics FINAL PROJECT REPORT

Milestones 1–4 | Weeks 1–8 | Python + Power BI



Figure 1. Supportlytics Power BI dashboard – Overview page.

Presented by Srikanth K

This report consolidates the supplied milestone presentations, analytical outputs, processed dataset and Power BI dashboard evidence into one final project document.

1. Executive Summary

Supportlytics is an end-to-end data analytics project developed to understand and optimize IT support operations. The project starts with ticket-level data and progressively converts it into business information: clean records, engineered metrics, exploratory visualizations, similarity scores, operational clusters, performance comparisons, geographic views and an interactive Power BI dashboard.

The project addresses practical support-management questions: where the workload is concentrated, which ticket types and categories dominate, how priority relates to resolution time, which queues carry the workload, which issue descriptions are similar, where unresolved high-priority tickets accumulate and how performance differs across countries and regions.

Final-data snapshot	Value
Processed records	11,409
Columns in final processed file	38
Average resolution time	120.60 hours
Average first-response time	36.39 hours
Average CSAT	3.01 / 5
Average similarity score	0.105
Average resolution duration	7.03 days
High/Urgent unresolved tickets	3,426

Dataset-version decision

Milestone 1 presentation material refers to a 200,000-row source dataset. The final processed CSV supplied for this report contains 11,409 records and 38 columns. All final quantitative findings in this document are calculated from the final processed CSV so that the report is internally consistent.

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Figure Index

Figure	Description
3–4	Ticket Type Distribution; Ticket Volume by Priority
5–6	Top Categories; Queue Distribution
7–8	Similarity Score; Cluster Similarity
9–10	Cluster Volume; Cluster Performance
11–12	Resolution by Priority; Resolution by Country
13–14	Resolution by Queue; High-Priority Unresolved
15–16	Regional Concentration; Regional-Priority Heatmap
17–22	Power BI Overview, Performance, Geography, Clusters, Ticket Analysis and Insights

2. Project Introduction and Problem Definition

3.1 Background

IT support teams receive tickets covering incidents, requests, product questions, account issues, payment issues, technical problems and other service needs. A ticket is more than a single record: it contains operational signals about urgency, customer context, response speed, resolution effort, queue ownership and service quality.

The project therefore treats the support-ticket dataset as an operational analytics source. The objective is to move from simple counting toward questions that can support decisions: which workloads require staffing, where delays occur, which issue families are repetitive, and where automation or standardization can improve efficiency.

3.2 Problem Statement

- Long resolution times can reduce service efficiency and customer satisfaction.
- High ticket backlog can hide critical cases that need immediate attention.
- Repeated issue descriptions create opportunities for knowledge-base articles, templates and automation.
- Uneven workload across queues can lead to inefficient resource allocation.
- Priority labels alone may not explain resolution performance; queue, type, category and geography must be considered together.

3.3 Project Goal

The goal is to optimize IT support team performance using analytics by building a reusable pipeline that cleans ticket data, derives operational metrics, discovers patterns and presents the results through an interactive Power BI dashboard.

3.4 Analytical Questions

- What types of tickets create the largest workload?
- How balanced is the priority distribution?
- Which categories and queues receive the most tickets?
- Which ticket descriptions are similar enough to support grouping?
- Which clusters have high workload or high resolution effort?
- Which priority, country, region and queue combinations show slower resolution?
- How many high/urgent tickets remain unresolved?
- How can all findings be presented to management through one dashboard?

3. Objectives, Scope and Milestone Progress

4.1 Objectives

- Load and understand the customer-support ticket dataset.
- Clean missing values, dates and analytical fields.
- Create Resolution_Duration_Days, Priority_Score, Type, Queue, Similarity_Score, Cluster_ID, Cluster_Name and Country.
- Perform descriptive and comparative exploratory data analysis.
- Use TF-IDF and cosine similarity to quantify related ticket descriptions.
- Use KMeans clustering to create five operationally meaningful issue groups.
- Compare resolution time by priority, country and support queue.
- Identify unresolved High/Urgent tickets by ticket type.
- Analyze regional ticket concentration and priority distribution.
- Build and document a six-page Power BI dashboard.

4.2 Milestone Progress

Milestone	Modules	Work actually completed
1 – Weeks 1–2	Modules 1–2	Dataset acquisition, schema review, missing-value checks, cleaning, date conversion, type/priority/queue logic and processed-data export.
2 – Weeks 3–4	Modules 3–4	EDA, category/queue analysis, TF-IDF text vectorization, cosine similarity, KMeans clustering, cluster naming and business interpretation.
3 – Weeks 5–6	Modules 5–6	Resolution-time analysis, country feature engineering, high-priority unresolved analysis, regional concentration and heatmap/geographic analysis.
4 – Weeks 7–8	Module 7	Power BI architecture, six analytical pages, KPIs, slicers, navigation, dashboard screenshots and executive insights.

Milestone continuity

The project is cumulative: Milestone 2 consumes the processed data from Milestone 1; Milestone 3 uses the engineered similarity, cluster and geographic fields; Milestone 4 integrates those outputs into Power BI.

4. Methodology and Technology Stack

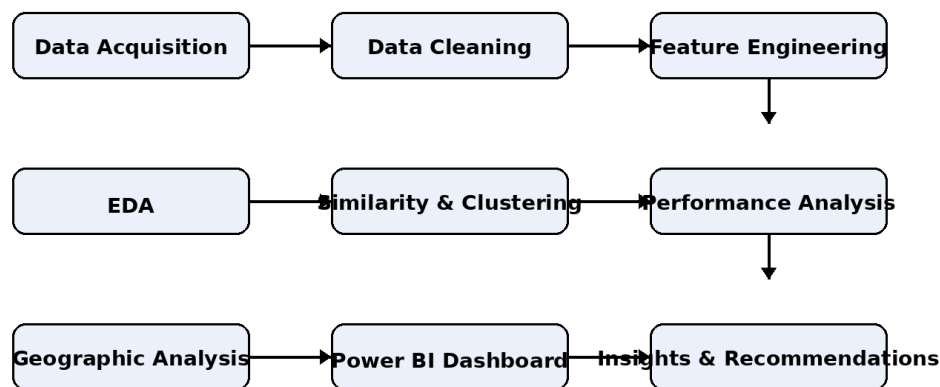


Figure 2. Project workflow from dataset preparation to dashboard and recommendations.

5.1 End-to-End Workflow

The workflow follows seven analytical stages. Data acquisition establishes the working dataset. Data cleaning makes the fields reliable. Feature engineering converts raw values into metrics that can be grouped and compared. Exploratory analysis identifies distributions and workload patterns. Similarity and clustering transform unstructured issue descriptions into operational groups. Performance and geographic analysis quantify where the support process behaves differently. Finally, Power BI combines the results into an interactive reporting layer.

5.2 Technology Stack

Technology	How it was used in Supportlytics
Python	Main programming language for the analytical pipeline.
Pandas	CSV loading, filtering, grouping, crosstabs, feature engineering and exports.
NumPy	Numerical calculations and supporting transformations.
Matplotlib	Bar charts, horizontal comparisons and performance plots.
Seaborn	Heatmap visualization and statistical-style graphics.
Scikit-learn	TF-IDF vectorization, cosine similarity and KMeans clustering.
Plotly	Geographic visualization capability in the milestone workflow.
Google Colab / Jupyter	Notebook execution, iterative analysis and milestone evidence.
Power BI	Interactive KPI reporting, slicers, navigation and management dashboard.

5.3 Analytical Design Principle

The project separates preparation from reporting. Python creates the analytical dataset and evidence, while Power BI acts as the consumption layer. This makes the workflow reusable: the processed table can be refreshed or replaced while the analytical logic and dashboard structure remain conceptually the same.

5. Dataset Understanding and Data Dictionary

6.1 Final Dataset

The final processed CSV contains 11,409 rows and 38 columns. The data combines original ticket/customer fields with engineered variables used specifically for this project. The final file is the principal source for the numerical findings in this report.

6.2 Logical Data Dictionary

Data area	Important columns	What they contribute
Ticket identity	ticket_id, product, issue_description, resolution_notes	Identify the case and describe the issue/resolution.
Classification	category, priority, status, Type, Queue	Define workload type, urgency and ownership.
Timing	ticket_created_date, ticket_resolved_date, first_response_time_hours, resolution_time_hours	Measure response and resolution efficiency.
Customer context	customer_age, customer_gender, subscription_type, customer_tenure_months, previous_tickets, customer_segment	Provide customer and account context.
Experience	customer_satisfaction_score, language, preferred_contact_time	Measure customer experience and communication context.
Technical context	operating_system, browser, payment_method, channel	Explain platform and contact-channel patterns.
Derived analytics	Resolution_Duration_Days, Priority_Score, Similarity_Score, Cluster_ID, Cluster_Name	Enable performance, similarity and cluster analysis.
Geography	region, Country	Enable regional and country comparisons.

6.3 Data Granularity

The analytical grain is one row per support ticket. Aggregations such as ticket count, average resolution time, average CSAT and similarity are therefore calculated by grouping these ticket-level observations.

6.4 Sensitive Fields

Customer name and email exist in the source but are intentionally excluded from report tables and screenshots because they are not required for operational analytics.

6. Data Preprocessing and Quality Checks

7.1 Loading and Profiling

The Milestone 1 notebook loads the CSV with Pandas, inspects shape and column names, checks data types, reviews missing values and performs descriptive analysis. This establishes whether the raw structure is suitable for downstream grouping and visualization.

```
import pandas as pd
df = pd.read_csv("customer_sp_ticket.csv")
print(df.shape)
df.info()
df.isnull().sum()
```

7.2 Missing Values

The milestone analysis identifies Browser as the main field requiring imputation. The notebook fills missing Browser values using the mode. The final processed file supplied for this report contains completed Browser values.

7.3 Date Handling

```
df['ticket_created_date'] = pd.to_datetime(df['ticket_created_date'], errors='coerce')
df['ticket_resolved_date'] = pd.to_datetime(df['ticket_resolved_date'], errors='coerce')
df['Resolution_Duration_Days'] = (
    df['ticket_resolved_date'] - df['ticket_created_date']
).dt.days
```

In the final processed dataset, Resolution_Duration_Days has a minimum of 0 days, a maximum of 14 days, an average of 7.03 days and a median of 7 days. No negative durations are present.

7.4 Validation

- Check exact duplicates during cleaning.
- Standardize labels used for grouping.
- Validate derived fields after transformation.
- Confirm missing-value counts after imputation.
- Export the processed analytical dataset for the next milestones.

7. Feature Engineering and Derived Logic

8.1 Priority Score

Priority is transformed from a text label into an ordered numeric score so that it can be used consistently in calculations and analytical comparisons.

Priority	Score	Interpretation
Low	1	Lowest urgency
Medium	2	Normal operational priority
High	3	Elevated urgency
Urgent	4	Highest urgency

```
priority_map = {'Low':1, 'Medium':2, 'High':3, 'Urgent':4}
df['Priority_Score'] = df['priority'].map(priority_map)
```

8.2 Ticket Type Mapping

Categories in the project logic	Derived Type
Feature Request; Subscription Cancellation; Refund Request; Account Suspension	Request
Performance Issue; Bug Report; Payment Problem	Problem
Login Issue; Security Concern; Data Sync Issue	Incident

8.3 Queue Mapping

The project maps support categories into operational queues. In the final data, General Support and Finance Team are the dominant queue labels. This field allows workload and resolution performance to be compared by ownership.

8.4 Why These Features Matter

- Priority_Score makes urgency measurable.
- Type turns individual categories into higher-level operational workload classes.
- Queue connects tickets to support ownership.
- Resolution_Duration_Days provides a direct duration measure.
- Similarity and cluster fields turn text descriptions into comparable issue groups.
- Country extends regional data into a country-level comparison for the project dashboard.

8. Exploratory Data Analysis – Ticket Type and Priority

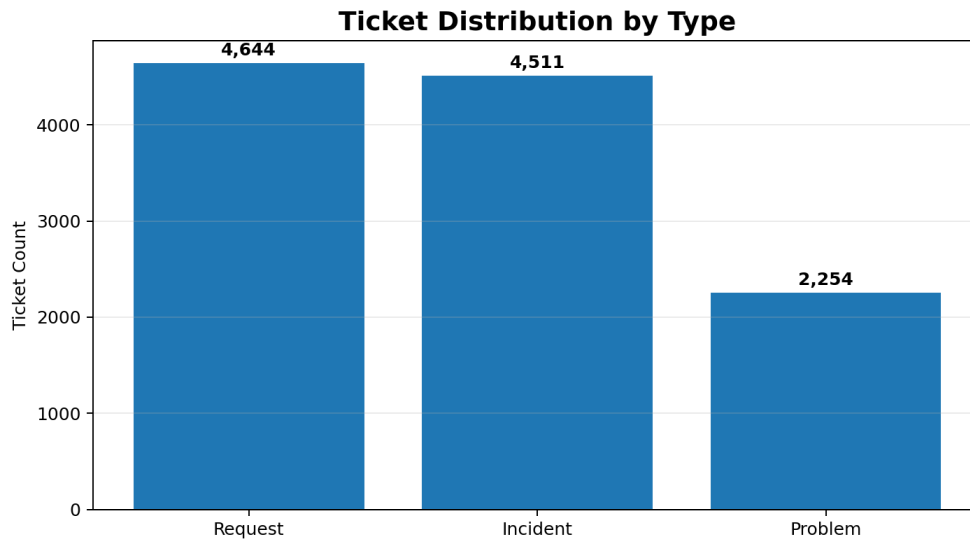


Figure 3. Ticket distribution by derived ticket type.

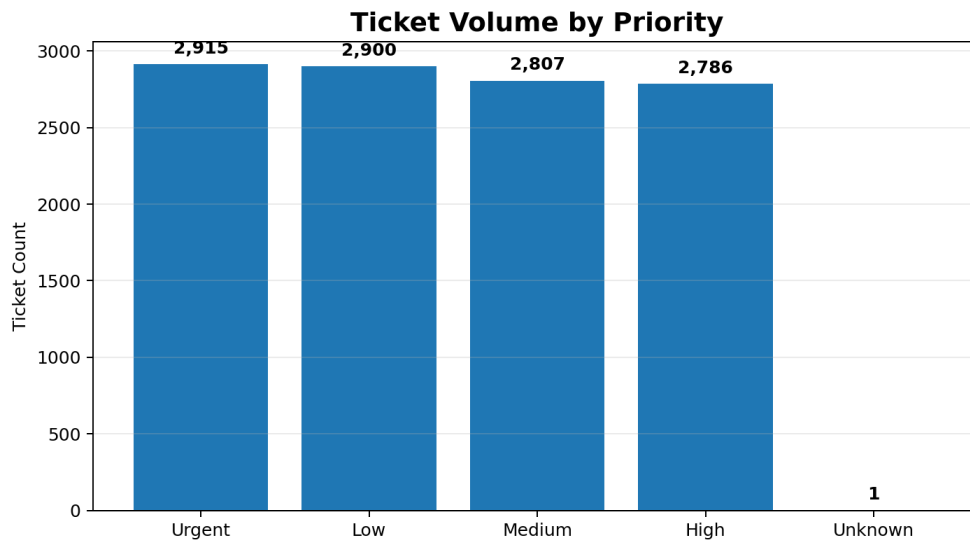


Figure 4. Ticket volume by priority.

9.1 Ticket Type Findings

Request tickets are the largest type with 4,644 records (40.7%). Incident tickets contribute 4,511, while Problem tickets contribute 2,254. The distribution indicates that support workload is driven mainly by requests and incidents rather than problem-management cases.

9.2 Priority Findings

Urgent has the highest count at 2,915, followed by Low (2,900), Medium (2,807) and High (2,786). The relatively balanced distribution means staffing cannot be planned around only one priority level.

9.3 Business Meaning

The type distribution informs workforce specialization, while the priority distribution informs triage and escalation. Together, they provide a baseline for deciding whether the support operation should focus on request automation, incident response or problem-management capacity.

9. Exploratory Data Analysis – Category, Queue and Workload

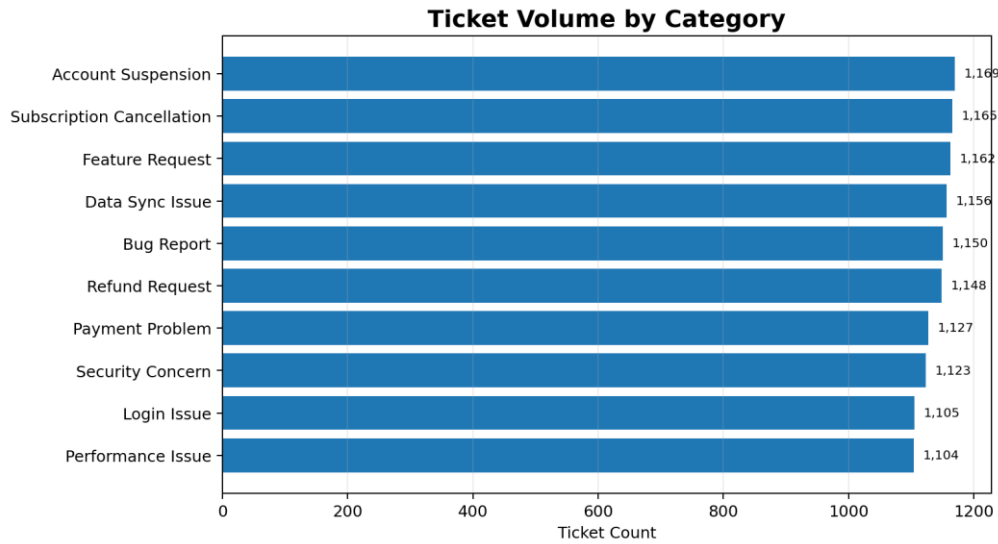


Figure 5. Ticket volume by issue category.

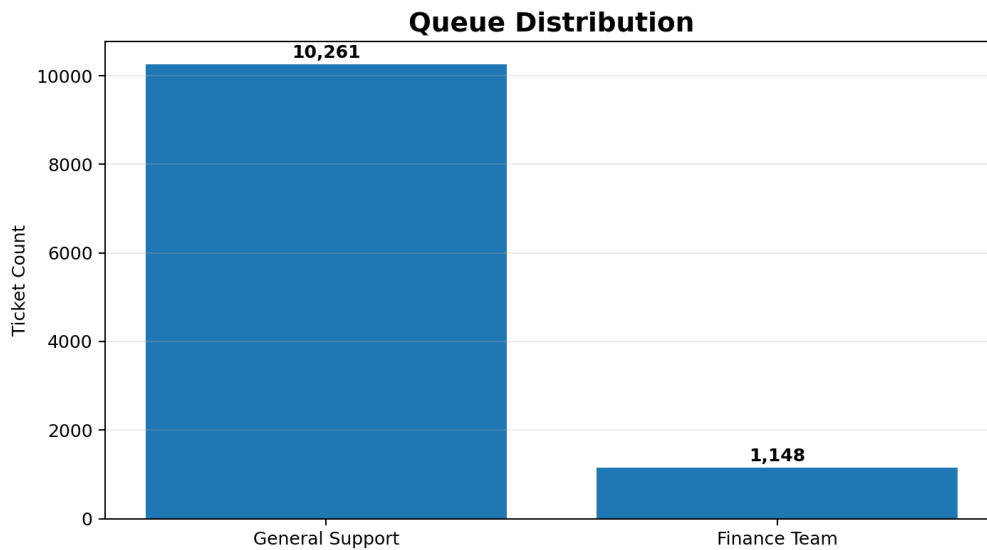


Figure 6. Ticket volume by support queue.

10.1 Category Analysis

The final dataset contains 10 issue categories. Account Suspension has 1,169 tickets, followed by Subscription Cancellation with 1,165. The category distribution is comparatively broad, indicating that the support operation must manage multiple issue families rather than one dominant problem.

10.2 Queue Analysis

General Support handles 10,261 tickets (89.9%), while Finance Team handles 1,148 (10.1%). This is the clearest workload imbalance in the final dataset.

Operational implication

Queue volume should be considered alongside resolution time. A queue with more tickets is not automatically inefficient; it may simply own more of the organization's demand. The appropriate question is whether workload and service performance are balanced.

10. Similarity Analysis

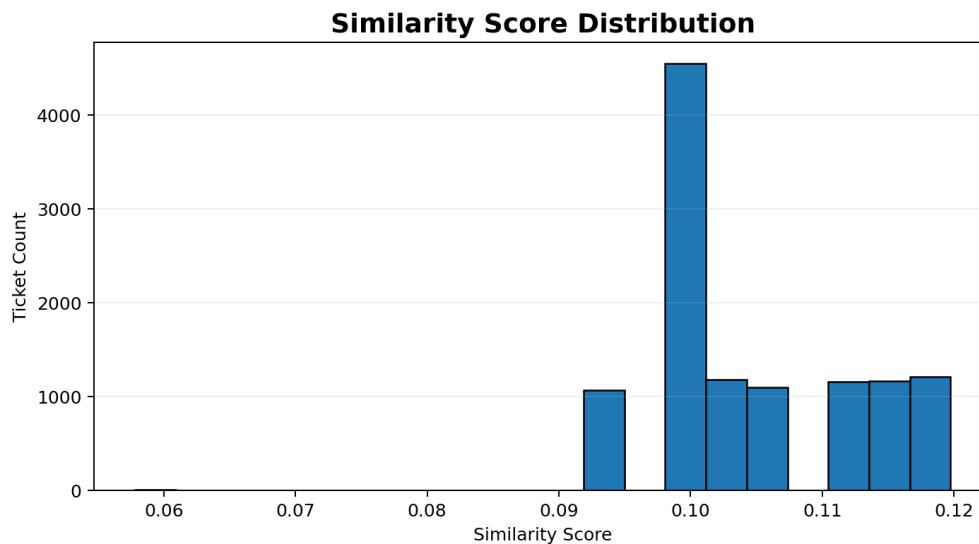


Figure 7. Distribution of similarity scores generated from issue descriptions.

11.1 Text Representation

Milestone 2 converts issue descriptions into numerical TF-IDF vectors. TF-IDF gives higher weight to words that are important within a ticket but less common across the full collection. This creates a numerical representation that can be compared between tickets.

```
from sklearn.feature_extraction.text import TfidfVectorizer
vectorizer = TfidfVectorizer(stop_words='english', max_features=5000)
X = vectorizer.fit_transform(df['issue_description'].fillna(''))
```

11.2 Cosine Similarity

Cosine similarity compares the angle between two TF-IDF vectors. A higher score indicates that two ticket descriptions have more similar term patterns. The project summarizes similarity scores and uses them as an input to clustering and reporting.

```
from sklearn.metrics.pairwise import cosine_similarity
similarity = cosine_similarity(X)
similarity_score = similarity.mean(axis=1)
df['Similarity_Score'] = similarity_score
```

The final file has similarity scores from 0.058 to 0.120, with a mean of 0.105. The distribution is concentrated in a relatively narrow range, indicating repeated language patterns across the support descriptions.

11. Cluster Formation and Interpretation

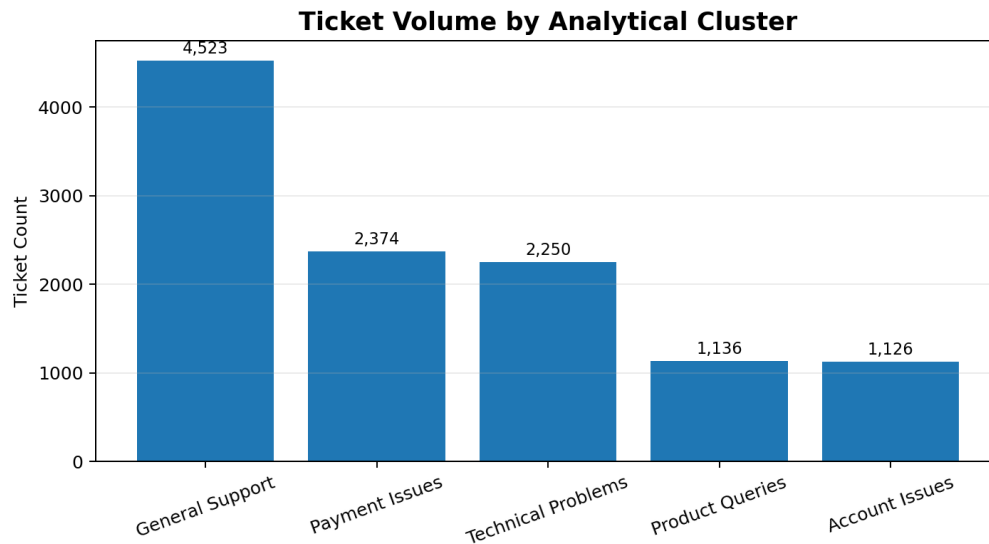


Figure 8. Ticket volume by business-oriented cluster.

12.1 KMeans Clustering

The project uses KMeans to group tickets according to their text-vector representation. Five clusters are created and then translated into business-oriented names so that the output can be understood by support stakeholders.

```
from sklearn.cluster import KMeans
kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)
df['Cluster_ID'] = kmeans.fit_predict(X)
```

12.2 Cluster Names

Cluster name	Business interpretation
Technical Problems	Technical and system-related issue patterns.
General Support	Broad support questions and routine service needs.
Payment Issues	Payment, billing and transaction-related patterns.
Product Queries	Questions about products or product functionality.
Account Issues	Account access, account status and related requests.

12.3 Why Clustering Was Included

Clustering extends the project beyond simple category counts. It can reveal groups of tickets that use similar language even when the original category labels do not fully describe their relationship. These groups can support routing, knowledge-base design and future automation.

12. Cluster Performance Analysis

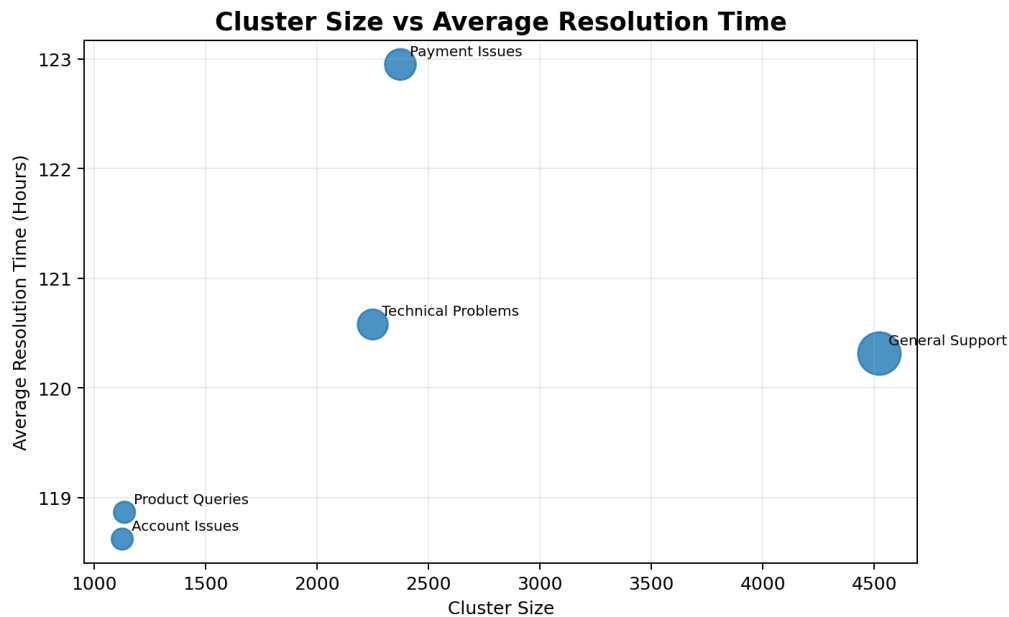


Figure 9. Cluster size versus average resolution time.

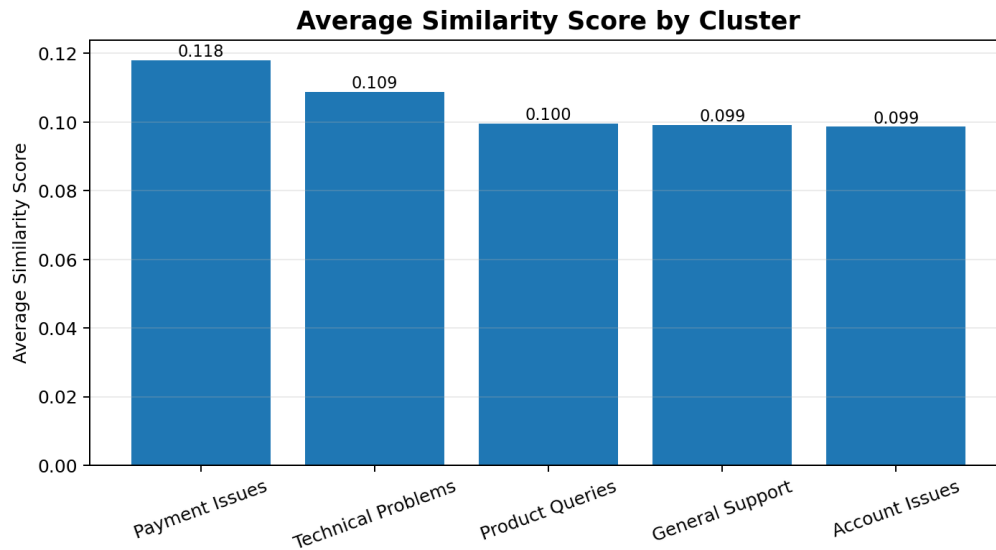


Figure 10. Average similarity score by cluster.

Cluster	Tickets	Avg. resolution (h)	Avg. similarity
General Support	4,523	120.31	0.099
Payment Issues	2,374	122.95	0.118
Technical Problems	2,250	120.58	0.109
Product Queries	1,136	118.87	0.100
Account Issues	1,126	118.62	0.099

Payment Issues has the highest cluster-level average resolution time at 122.95 hours and the highest mean similarity score at 0.118. This combination makes Payment Issues a strong candidate for process standardization: the descriptions are relatively consistent, yet the average handling time is comparatively high.

General Support is the largest cluster, so even moderate inefficiency in this group can have a large operational effect because of its workload size. Technical Problems also represents a substantial workload and can benefit from structured troubleshooting and reusable resolution steps.

13. Resolution-Time Performance

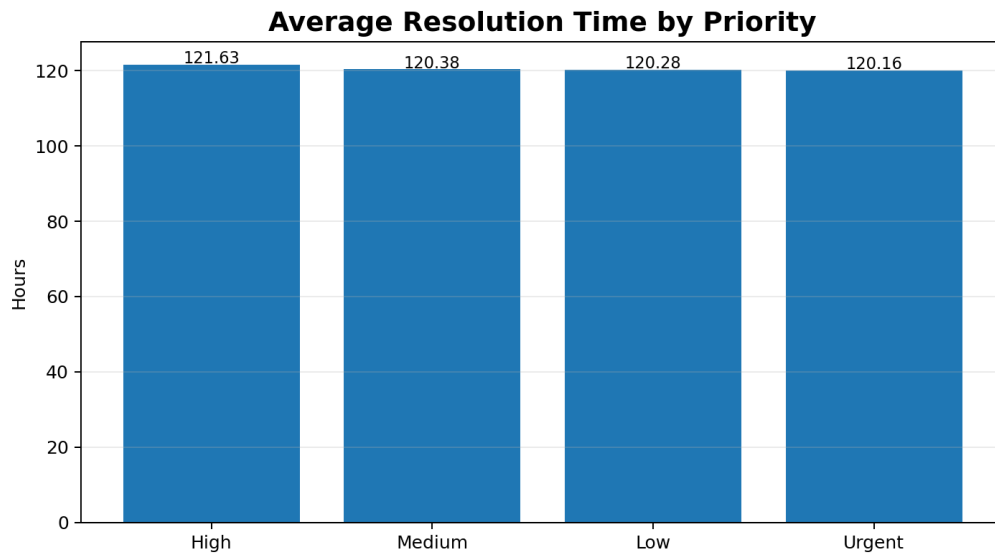


Figure 11. Average resolution time by priority.

14.1 Priority Performance

Average resolution time is tightly grouped around 120 hours: High 121.63, Medium 120.38, Low 120.28 and Urgent 120.16 hours.

14.2 Timing KPIs

KPI	Final-data value	Interpretation
Average Resolution Time	120.60 h	Typical completion effort at ticket level.
Average First Response Time	36.39 h	Average time before initial response.
Average Resolution Duration	7.03 days	Calendar-day view of ticket duration.
Median Resolution Duration	7 days	Middle ticket duration.
75th percentile duration	11 days	Upper-quarter workload indicator.
Average CSAT	3.01/5	Customer experience indicator.

14.3 Interpretation

The near-flat priority pattern is important. A High or Urgent label does not produce a dramatically different average resolution time in the final dataset. This suggests that operational constraints such as queue capacity, issue complexity, routing and process steps may be as important as priority itself.

14. Country and Queue Performance

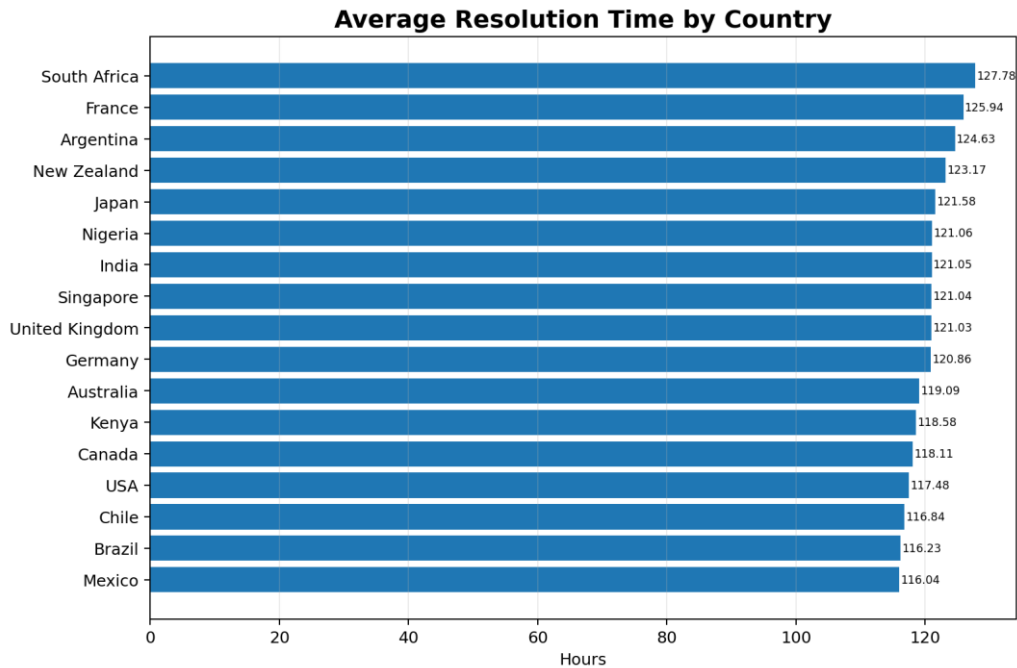


Figure 12. Average resolution time by country.

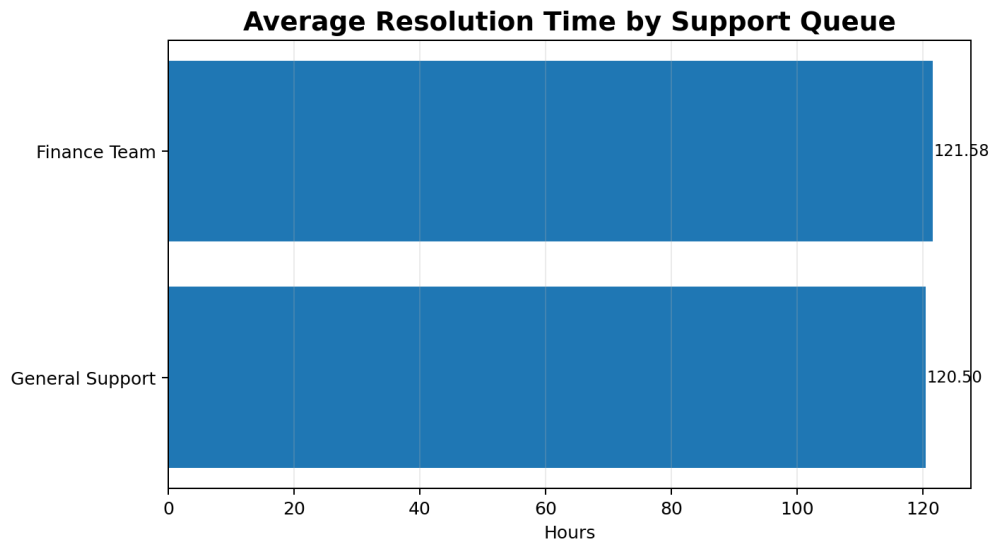


Figure 13. Average resolution time by support queue.

15.1 Country Comparison

Among named countries, South Africa has the highest average resolution time (127.78 h), while Mexico has the lowest (116.04 h). This spread provides a benchmarking opportunity for support leadership.

15.2 Queue Comparison

Finance Team averages 121.58 hours and General Support averages 120.50 hours. Their resolution averages are close even though General Support carries much more ticket volume.

15.3 Management Use

Country-level variation can be used to investigate staffing, time-zone coverage, process maturity or issue mix. Queue-level comparison can be used to identify whether a team is slower because of workload, complexity or process design.

15. High-Priority Unresolved Analysis

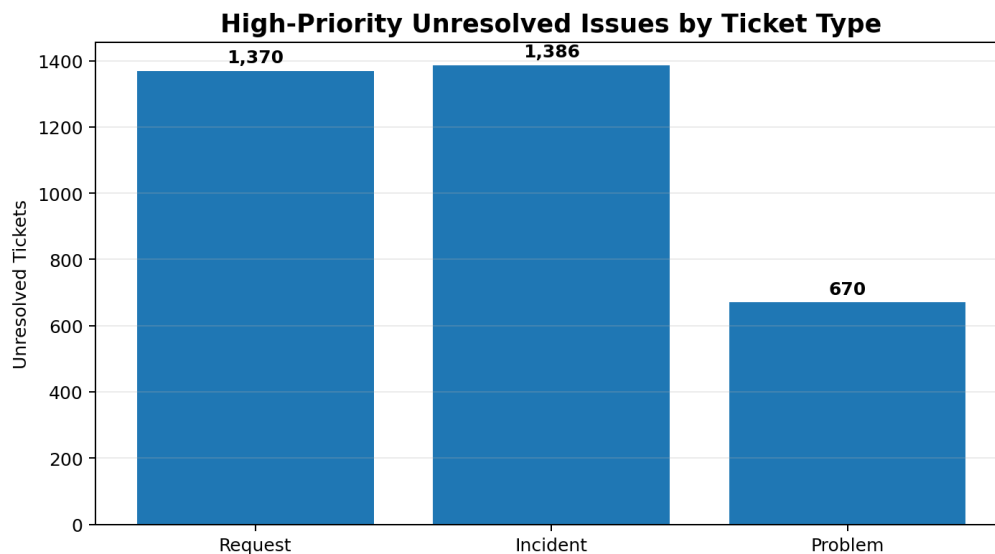


Figure 14. High and urgent unresolved tickets by ticket type.

16.1 Critical Workload

There are 3,426 tickets with High or Urgent priority that are not in Resolved or Closed status. By type, Request contributes 1,370, Incident contributes 1,386, and Problem contributes 670.

16.2 Why It Matters

- High-priority unresolved tickets represent operational risk because urgency has not yet translated into closure.
- Incident and Request cases form the largest critical unresolved groups and should be visible to supervisors.
- An aging view would be more actionable than a simple count because it can distinguish newly opened tickets from long-running cases.
- The Power BI dashboard can provide the filtering layer needed to isolate these cases by queue, country, region, category and type.

16.3 Recommended Control

Create a daily critical-ticket review containing ticket age, priority, queue, category, current status and escalation flag. Add an automated threshold such as 'High/Urgent open beyond X hours' once a production SLA definition is available.

16. Regional Ticket Concentration

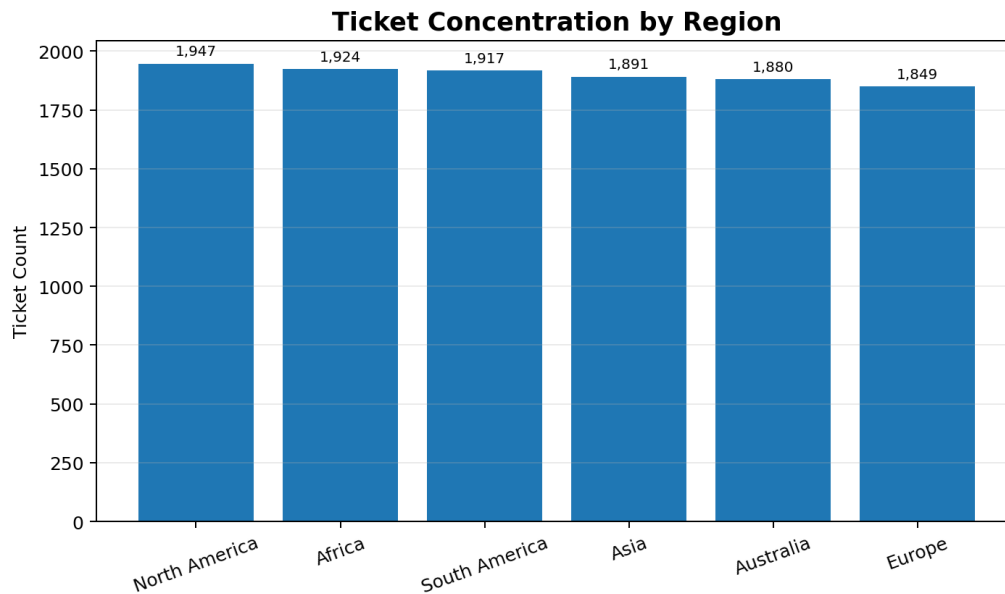


Figure 15. Ticket concentration by region.

17.1 Regional Volume

Region	Tickets	Share of final dataset
North America	1,947	17.07%
Africa	1,924	16.86%
South America	1,917	16.80%
Asia	1,891	16.57%
Australia	1,880	16.48%
Europe	1,849	16.21%
Unknown	1	0.01%

17.2 Findings

North America records 1,947 tickets, the highest among named regions. Africa records 1,924, South America 1,917, Asia 1,891, Australia 1,880 and Europe 1,849. The differences are modest, indicating that demand is distributed globally rather than being concentrated in a single region.

17.3 Regional Performance

North America has the lowest named-region average resolution time (117.19 h), while Europe has the highest (122.60 h). This provides a useful starting point for regional benchmarking, but the engineered country field and issue mix should be considered before drawing causal conclusions.

17. Geographic Heatmap and Country Mapping

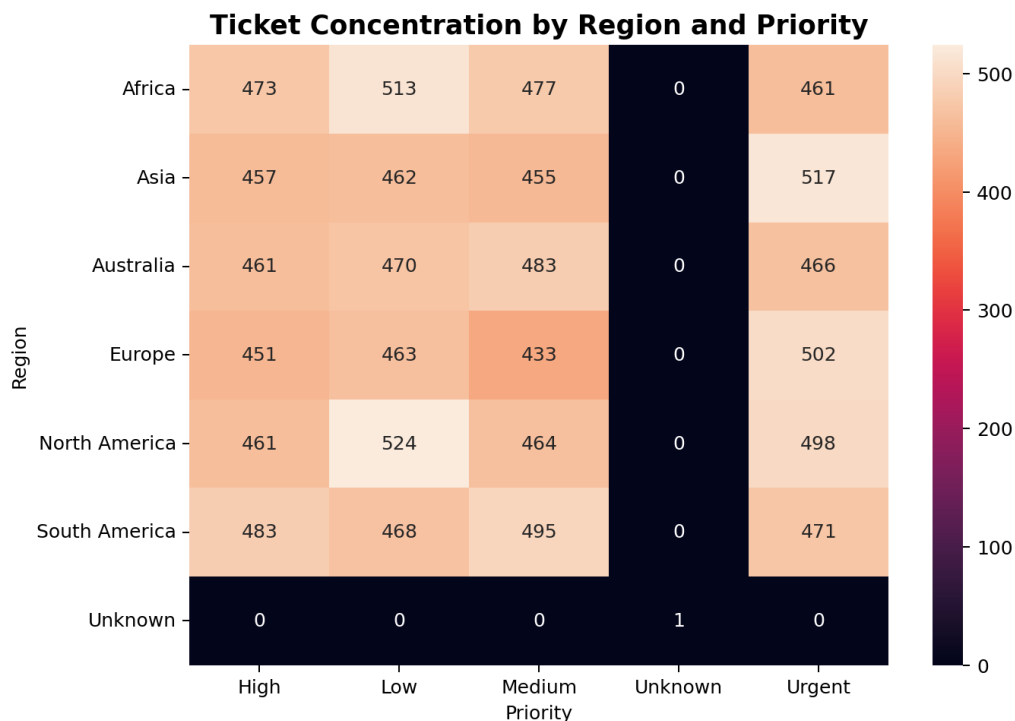


Figure 16. Ticket concentration by region and priority.

18.1 Heatmap Interpretation

The heatmap shows the joint distribution of region and priority. High, Medium, Low and Urgent tickets occur across all named regions. North America has the highest Low count (524), while Asia has the highest Urgent count (517) among the named regions. The view supports regional workload planning by priority.

18.2 Country Feature Engineering

The original project dataset did not contain a country field suitable for the required geographic analysis. Milestone 3 therefore created Country from Region using representative mappings: North America → USA, Canada, Mexico; South America → Brazil, Argentina, Chile; Europe → Germany, France, United Kingdom; Asia → India, Japan, Singapore; Africa → South Africa, Kenya, Nigeria; Australia → Australia, New Zealand; Unknown → Unknown.

18.3 Limitation of This Mapping

Use of Country in this report

Country is a project-engineered analytical feature created from region membership. It should be used for comparative visualization and dashboard demonstration, not as precise customer geolocation. A production implementation should source country directly from a trusted customer/account location field.

18. Power BI Dashboard Architecture

19.1 Dashboard Pages

Page	Purpose	Key content
Overview	Immediate support-health snapshot	Total tickets, resolution time, FRT, SLA/CSAT indicators, status, priority, type, categories, queues and trends.
Performance	Resolution efficiency	Priority, year, queue, type and country comparisons.
Geography	Regional workload	Countries, regions, ticket concentration, priority distribution and regional resolution.
Clusters	Issue similarity	Similarity, cluster volume, cluster size and resolution performance.
Ticket Analysis	Detailed demand composition	Ticket type, category, queue, priority and category × priority.
Executive Insights	Decision support	Top performers, fastest/slowest areas, critical open work and major clusters.

19.2 Interaction Design

- Navigation buttons provide a consistent path between analytical pages.
- Slicers enable comparative analysis instead of static reporting.
- KPI cards provide context before users interpret charts.
- The pages follow a narrative from overall health to detailed causes and management actions.

19.3 KPI Definitions Used in the Project

KPI	Project meaning
Average Resolution Time	Mean resolution_time_hours for the selected filter context.
First Response Time	Mean first_response_time_hours.
Total Tickets	Count of ticket records in the current filter context.
CSAT	Average customer_satisfaction_score.
Cluster Similarity Index	Average Similarity_Score for the selected cluster/filter context.
High Priority Open Tickets	Count of High/Urgent records not in final closed/resolved states.

Dashboard role

Milestone 4 converts notebook-based findings into an interactive decision-support interface, allowing managers to filter the same analytical results by operational dimensions.

19. Dashboard – Overview and Performance



Figure 17. Power BI Overview page.

Overview Page

The Overview page acts as the dashboard entry point. It brings together the headline support KPIs and the major workload dimensions. Ticket status, priority, type, monthly trend, top categories and queue performance can be reviewed before moving into deeper analysis.

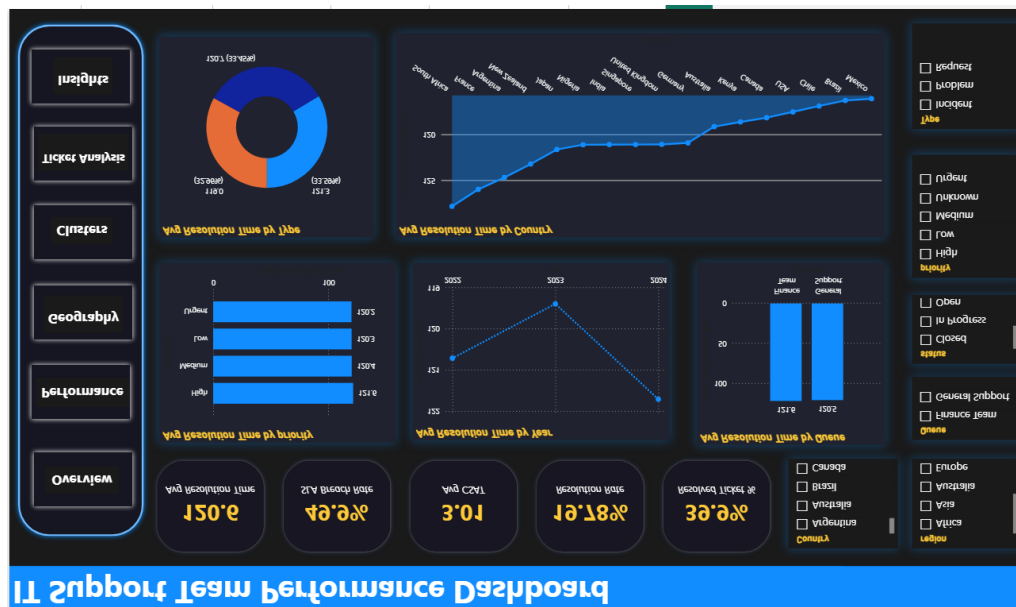


Figure 18. Power BI Performance Analysis page.

Performance Page

The Performance page focuses on service efficiency. It combines resolution-time comparisons with queue, ticket-type, country and time-based views. This page operationalizes the Milestone 3 performance analysis and provides the user with filters for comparative investigation.

What a manager can answer

- Is resolution time changing?
- Which priority or ticket type is slower?
- Which queue is handling the workload?
- Which country/region can act as a benchmark?
- Where should performance investigation begin?

21. Dashboard – Geography and Clusters

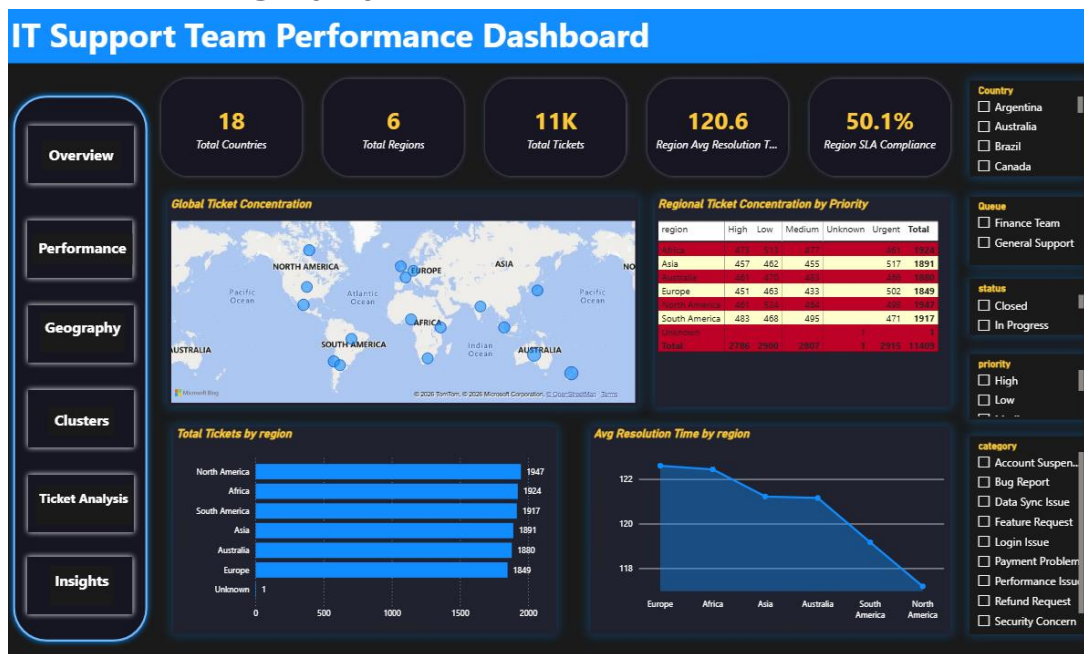


Figure 19. Power BI Geographic Analysis page.

Geography Page

The Geography page connects regional and country-level demand with performance. It supports map-based exploration, regional priority analysis, ticket concentration and resolution-time comparison.

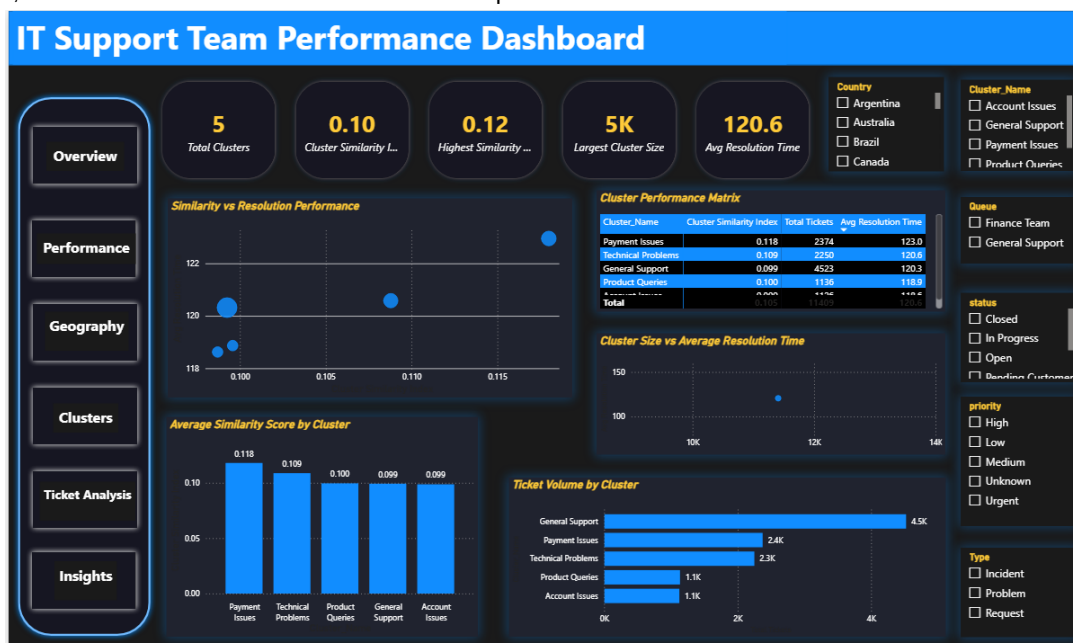


Figure 20. Power BI Cluster Analysis page.

Clusters Page

The Clusters page translates the Milestone 2 text analytics into a management view. It allows users to compare cluster size, similarity and resolution performance, helping identify issue families that are repetitive, large or operationally expensive.

22. Dashboard – Ticket Analysis and Executive Insights

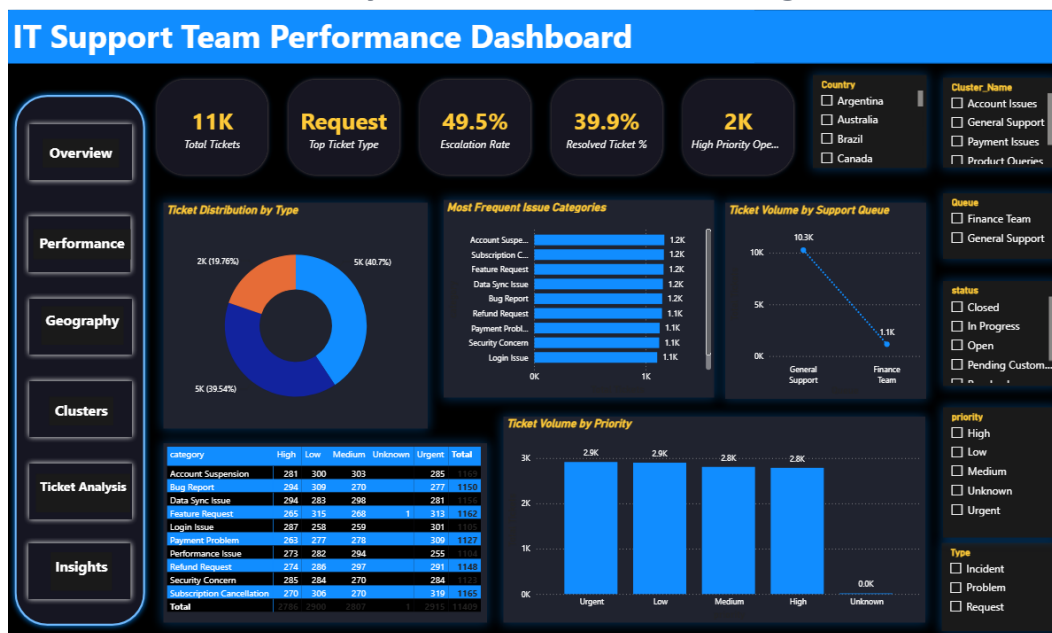


Figure 21. Power BI Ticket Analysis page.

Ticket Analysis Page

This page provides detailed workload composition: ticket type, category frequency, queue volume, priority distribution and category × priority patterns. It is intended for supervisors who need to move from a KPI to the specific workload driving the KPI.

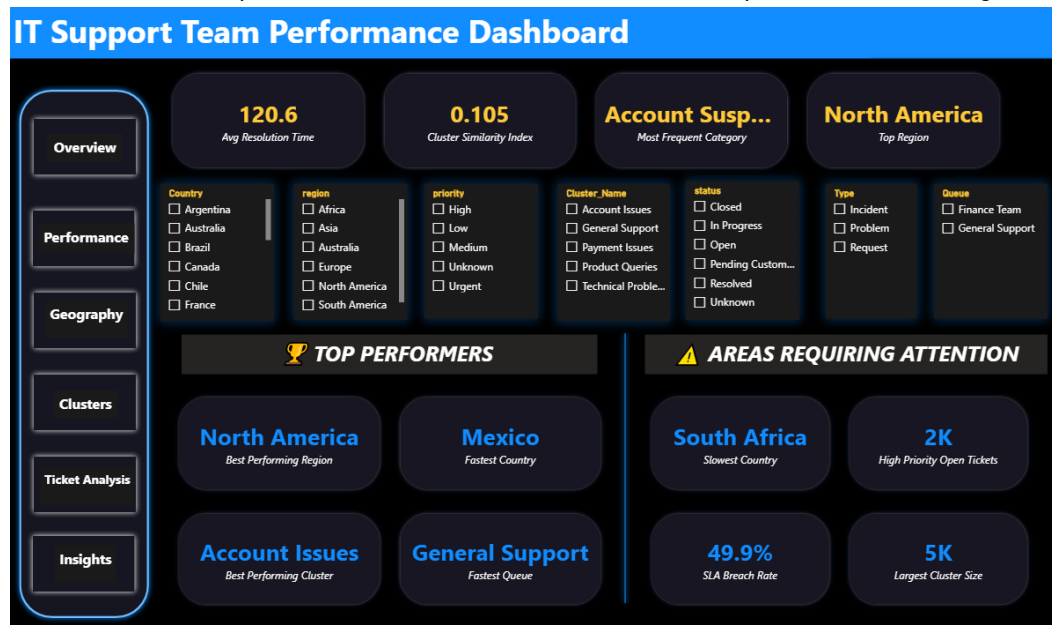


Figure 22. Power BI Executive Insights page.

Executive Insights Page

The final page summarizes top-performing regions, fastest country/queue, best-performing cluster and areas requiring attention such as the slowest country, high-priority open work, SLA risk and large clusters. It is designed to reduce the time required to translate analytical visuals into management actions.

23. Key Findings and Recommendations

23.1 Consolidated Findings

Finding	Evidence from final analysis	Management meaning
Request is largest ticket type	4,644 tickets	Prioritize request automation/self-service.
General Support dominates workload	10,261 tickets	Review capacity and routing.
Priority resolution is nearly flat	All priority averages are near 120 h	Investigate complexity, queues and process bottlenecks.
Critical unresolved workload exists	3,426 High/Urgent unresolved	Use aging and escalation controls.
Payment Issues is highly consistent	Similarity 0.118	Good candidate for standard workflows.
Payment Issues is slower	122.95 h average	Investigate billing/payment process bottlenecks.
Regional demand is broad	Named regions range 1,849–1,947 tickets	Plan globally rather than for one hotspot.
Country performance varies	116.04–127.78 h	Use benchmarks and investigate process differences.

23.2 Recommendations

1. Create an operational aging view for High and Urgent tickets and review it daily.
2. Review General Support capacity because it owns the majority of the final workload.
3. Standardize Payment Issues troubleshooting and create reusable resolution templates.
4. Use the fastest-performing regions/countries as internal benchmarks, while checking issue mix before comparing teams.
5. Track first response and resolution together to distinguish acknowledgement speed from completion speed.
6. Use recurring categories and high-similarity clusters to prioritize knowledge-base articles and self-service.
7. Add production SLA thresholds and automated alerts once official SLA definitions are available.
8. Extend the dashboard with predictive resolution-time and escalation-risk models in a future phase.

24. Limitations, Future Scope, Conclusion and Appendix

24.1 Limitations

- The project uses a supplied analytical support-ticket dataset; it should not be interpreted as live production service-management data.
- Milestone material and final processed data represent different dataset versions; this report uses the final CSV for numerical consistency.
- Country is engineered from region and is therefore not exact customer geolocation.
- The current workflow is primarily batch analytics rather than real-time streaming.
- The dashboard screenshots represent the completed dashboard interface; some displayed KPI values may reflect the dashboard's capture-time filter context.

24.2 Future Scope

- Connect Power BI to a production ITSM system for scheduled/real-time refresh.
- Create automated critical-ticket aging alerts.
- Build supervised models for resolution-time and escalation prediction.
- Use NLP sentiment and intent classification for smarter routing.
- Use similarity matching to recommend existing solutions and knowledge-base articles.
- Create role-specific views for agents, supervisors and executives.

24.3 Conclusion

Supportlytics demonstrates a complete analytics lifecycle for IT support operations. The project does not stop at descriptive charts: it creates derived operational features, applies text analytics and clustering, compares service performance across multiple dimensions, introduces geographic analysis and finally integrates the results into an interactive Power BI dashboard.

The final analytical evidence shows a workload dominated by Request tickets and General Support, resolution averages close to 120 hours across priority levels, a measurable population of 3,426 unresolved High/Urgent tickets, meaningful country-level resolution variation and a Payment Issues cluster that combines high similarity with comparatively high resolution time. These findings provide clear starting points for workload balancing, critical-ticket monitoring, process standardization and automation.

Final deliverables: Milestone 1–4 analyses and presentations, processed analytical CSV, completed six-page Power BI dashboard, dashboard visual evidence, and this final report.